CLASS 9 · CBSE · GANITA MANJARI PART 2 · FORMULA BANK

Chapter 12: Quadrilaterals

Every formula and result of the chapter on one sheet

1	Angle sum of a quadrilateral	360°
2	Convex quadrilateral	Every interior angle is less than 180°; the diagonals intersect
3	Parallelogram	Opposite sides are parallel
4	Properties of a parallelogram	Opposite sides equal, opposite angles equal, diagonals bisect each other
5	Adjacent angles of a parallelogram	Add up to 180°
6	Tests for a parallelogram	Opposite sides equal; opposite angles equal; diagonals bisect each other; one pair of sides equal and parallel
7	Midpoint Theorem	$PQ \parallel BC$ and $PQ = \frac{1}{2}BC$
8	Converse of the Midpoint Theorem	A line through the mid-point of one side, parallel to a second side, bisects the third side
9	Median	Joins a vertex to the mid-point of the opposite side
10	Centroid	The medians meet at one point, which divides each in the ratio $2:1$ from the vertex
11	Varignon parallelogram	Formed by the mid-points of the sides of any quadrilateral
12	Perimeter of the Varignon parallelogram	Sum of the diagonals of the quadrilateral
13	Angle sum of an n-gon	$(n-2) \times 180^\circ$
14	Number of diagonals of an n-gon	$\frac{n(n-3)}{2}$